INTRODUCTION TO MACHINE LEARNING AND AI

Concepts, workflows, evaluation and real-world applications

Purpose: This sample source document is designed for a document-retrieval practical. It contains multiple related topics that can be extracted, split into chunks, embedded, and searched using a vector database.

1. What Is Artificial Intelligence?

Artificial intelligence (AI) is a field of computing concerned with building systems that perform tasks commonly associated with human intelligence. These tasks include understanding language, recognizing images, identifying patterns, making recommendations, and supporting decisions. An AI system uses data, algorithms, and computational resources to produce an output for a given task.

AI is a broad area that includes machine learning, deep learning, natural language processing, and computer vision. Not every AI system learns from data. Some systems follow rules written by developers, while machine-learning systems learn patterns from examples.

2. What Is Machine Learning?

Machine learning (ML) is a branch of AI that enables computers to learn patterns from data and use those patterns to make predictions or decisions. Instead of writing a separate rule for every possible situation, developers train a model using examples. The model adjusts its internal parameters during training to reduce the difference between its predictions and the expected results.

A typical machine-learning workflow includes defining the problem, collecting data, cleaning and preparing the data, selecting useful features, splitting the data, training a model, evaluating its performance, and deploying or monitoring the model.

3. Main Types of Machine Learning

Supervised learning uses labelled training examples. Each example contains input data and a known target output. The model learns a relationship between the inputs and targets, then uses that relationship to predict outputs for new examples. Classification predicts a category, such as whether an email is spam. Regression predicts a continuous numerical value, such as the expected price of a house.

Unsupervised learning works with data that does not have labelled target outputs. The objective is to discover structure or patterns in the data. Clustering groups similar data points together. Dimensionality reduction represents data using fewer features while attempting to preserve useful information.

Reinforcement learning trains an agent to make decisions by interacting with an environment. The agent takes an action, receives feedback in the form of a reward or penalty, and learns a strategy that aims to maximize cumulative reward over time. It is used in robotics, simulations, and sequential decision-making.

4. Data Preparation and Feature Engineering

Data preparation converts raw data into a form suitable for modelling. Common activities include handling missing values, removing duplicate records, correcting inconsistent formats, identifying outliers, and encoding categorical variables. Data should also be checked for leakage, which occurs when information that would not be available at prediction time enters the training process.

Feature engineering creates, transforms, or selects input variables that help a model learn. For example, a timestamp can be transformed into hour, day, or month features. Numerical features may be scaled so that variables with large ranges do not dominate certain algorithms. One-hot encoding represents a categorical value using a binary indicator for each category.

5. Training, Testing and Generalization

A dataset is often divided into training, validation, and test sets. The training set is used to fit model parameters. The validation set helps compare models and tune hyperparameters. The test set provides a final estimate of performance on data that was not used during model development.

Generalization is a model's ability to perform well on new, unseen examples. Overfitting occurs when a model learns details or noise specific to its training data and performs poorly on new data. Underfitting occurs when a model is too simple to capture important patterns. Regularization, cross-validation, appropriate model complexity, and more representative data can help improve generalization.

6. Model Evaluation Metrics

Evaluation metrics help determine whether a model is suitable for its intended task. For classification, accuracy is the proportion of all predictions that are correct. Precision measures the proportion of predicted positive cases that are actually positive. Recall measures the proportion of actual positive cases that the model correctly identifies. The F1-score is the harmonic mean of precision and recall.

Accuracy can be misleading when classes are imbalanced. For example, if only one percent of transactions are fraudulent, a model that predicts every transaction as legitimate can achieve high accuracy while detecting no fraud. In such cases, precision, recall, F1-score, and the confusion matrix provide additional insight.

For regression, common metrics include mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE). MAE reports the average absolute prediction error. MSE squares errors before averaging, giving larger errors more influence. RMSE is the square root of MSE and is expressed in the same units as the target variable.

7. Deep Learning and Neural Networks

Deep learning is a subfield of machine learning that uses neural networks with multiple layers. A neural network contains connected computational units called neurons. Each connection has a weight, and neurons apply an activation function to a weighted combination of their inputs. During training, the network adjusts its weights to reduce a loss function.

Convolutional neural networks (CNNs) are commonly used for image-related tasks because convolutional filters can learn local visual patterns. Recurrent neural networks (RNNs) and related architectures are designed to process sequences. Transformers use attention mechanisms to model relationships between elements in a sequence and are widely used in modern language models.

8. Natural Language Processing and Embeddings

Natural language processing (NLP) enables computers to process and work with human language. Common NLP tasks include text classification, sentiment analysis, named-entity recognition, translation, summarization, and question answering. Text is commonly cleaned and transformed into numerical features before being processed by a machine-learning model.

An embedding represents text as a dense numerical vector. Embedding models are designed so that texts with related meanings can have vectors that are close in vector space. This makes embeddings useful for semantic search, document retrieval, clustering, and recommendation systems.

9. Vector Search and Retrieval-Augmented Generation

A vector database stores numerical embeddings and supports similarity search. A document can be divided into smaller passages called chunks. Each chunk is converted into an embedding and stored with its original text and metadata, such as its source file or page number.

Cosine similarity compares the direction of two vectors. A cosine similarity closer to one indicates more similar directions. Some vector databases return cosine distance rather than similarity; in that representation, a smaller distance indicates a closer match. The distance metric should be selected explicitly when creating a collection if a particular metric is required.

Retrieval-Augmented Generation (RAG) combines document retrieval with text generation. First, the user's question is converted into an embedding. The system retrieves relevant document chunks from a vector database. Those chunks are then included as context in a prompt sent to a language model. The model uses the supplied context to formulate an answer. A well-designed RAG system should indicate when the retrieved context does not contain enough information rather than inventing an answer.

10. Responsible Use of AI

AI systems should be evaluated not only for accuracy but also for reliability, fairness, privacy, security, and explainability. Training data may contain errors or historical biases. A model can also behave differently when it encounters data that differs from its training distribution. Developers should document limitations, protect sensitive information, test representative cases, and monitor deployed systems.

11. Summary

Artificial intelligence is a broad field, and machine learning is one of its important branches. Machine-learning systems learn from data, while deep learning uses multilayer neural networks to model complex patterns. Data preparation and suitable evaluation are essential for building reliable models. In document-based AI applications, embeddings and vector search help retrieve relevant passages. RAG then supplies those passages to a language model so that its response can be grounded in a selected source.

Glossary

Term	Meaning
Artificial intelligence	Computing systems performing tasks associated with intelligent behaviour.
Machine learning	A branch of AI in which models learn patterns from data.
Feature	An input variable used by a model.
Label	The known target output in a supervised-learning example.
Embedding	A dense numerical vector representing data such as text.
Vector database	A system designed to store and search vector representations.
Cosine distance	A distance measure based on the angle between vectors; lower values indicate closer directions.
Chunk	A smaller passage created by splitting a larger document.
Metadata	Descriptive information stored alongside a document or vector.
RAG	Retrieval-Augmented Generation; retrieving source context before generating an answer.

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